R13

Code No: 126AP

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, February/March - 2022
DISTRIBUTED SYSTEMS
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1. Give five types of hardware resource and five types of data or software resource that can usefully be shared. Give examples of their sharing as it occurs in practice in distributed systems. [15]

2. Provide three specific and contrasting examples of the increasing levels of heterogeneity experienced in contemporary distributed systems and brief them. [15]

3. Explain the snapshot algorithm of Chandy and Lamport. [15]

4. Give an example execution of the ring-based algorithm to show that processes are not necessarily granted entry to the critical section in happened-before order. [15]

5. State the multicast communication principles and techniques in detail. [15]

6.a) Discuss the design issues and implementation of RPC in brief.
b) Evaluate the Case study: Sun RPC. [7+8]

7.a) Discuss about name services and domain name services.
b) State the applications of directory services. [8+7]

8.a) Explain how the two-phase commit protocol for nested transactions ensures that if the top-level transaction commits, all the right descendants are committed or aborted.
b) Explain how distributed deadlocks can be detected? [8+7]

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